

MICA MINECRAFT LAUNCHER

MCP Server Guide

Setup, architecture, security model and a complete reference for every tool and resource the launcher exposes over the Model Context Protocol.

DOCUMENT	docs/MCP_SERVER_GUIDE.pdf
APPLIES TO	Launcher 3.0 · MCP protocol revision 2025-06-18
FEATURE STATUS	Shipped and usable; phases 1–4 complete, phase 5 (background/tray presence) not started
DEFAULT STATE	Off. Nothing binds, no token is generated, no file is written until you switch it on
SOURCE	com.micatechnologies.minecraft.launcher.mcp (41 classes, 21 test classes)
REGENERATING	docs/pdf-generation-assets/mcp-server-guide → npm install && npm run build

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1 Overview

The launcher can host a Model Context Protocol server on loopback, so an MCP client — Claude Code, Cursor, Codex, or anything that speaks the protocol — can read your modpack library, explain why a game crashed, and, if you let it, install, edit, launch and uninstall packs. It is off by default, and everything it can do is gated twice: once by the settings you choose, and once by a prompt at the moment of the call.

1.1 What the MCP server does

MCP is a request/response protocol that lets a model call named tools and read named resources on a server. The launcher implements the server half. A connected client can:

- **Read the library.** Every pack the launcher knows about, installed or merely available, with its installed version, latest published version, whether an update is waiting, mod loader and stability flag.
- **Diagnose a crash.** `diagnose_launch_failure` returns the launcher's own crash analysis — category, summary, suggested fixes — alongside the raw report and the pack's version and loader, so the model answers "why did my game crash" without you pasting logs anywhere.
- **Read a manifest.** The full pack definition as JSON, either as a tool call or as a `mica://` resource a client can re-read cheaply.
- **Change things — only after a second opt-in.** Install from a URL, create a pack, fork one, add or remove a mod, uninstall, launch, stop the game.

Everything runs inside the launcher process that already owns the modpack folders, through the same code paths the GUI uses. There is no second implementation of installing a pack, and no separate process mutating files behind the launcher's back.

1.2 What it deliberately does not do

NON-GOAL	WHY
It is not an MCP client	The launcher never consumes an MCP server and never calls a model. No AI SDK is a dependency, and none is needed: the launcher is the thing being driven.
No network exposure	The listener binds <code>InetAddress.getLoopbackAddress()</code> and refuses to serve a non-loopback peer. There is no LAN mode and no remote authentication scheme to misconfigure.
No standalone server process	The launcher enforces single-instance because concurrent processes mutating the same modpack folders is unsafe — <code>SynchronizedFileManager</code> is in-process only. Every deployment shape therefore executes tools inside the one launcher process.
No credentials, ever	No tool, resource, error or log entry may return an access token, a client token, a refresh token, the account UUID, or the MCP bearer token itself. Section 4.6 covers how that is enforced structurally rather than by convention.

NON-GOAL	WHY
No MCP SDK	The server surface is small and the protocol is JSON-RPC 2.0. It is hand-rolled against the Gson already in the tree rather than pulling Jackson and Reactor into a fat JAR that already carries JavaFX, JNA, Lanterna and three RGB vendor SDKs.

1.3 Implementation status

The feature is shipped and usable. What follows is an honest ledger of where it stands, so this document is not read as describing more than exists.

COMPLETE

- JSON-RPC codec, method dispatch, error codes
- Loopback HTTP transport + admission checks
- Session model, registry and cap
- Approval engine, per-tool policies, session grants
- Three-outcome consent dialog with a deadline
- 7 read-only tools, 8 state-changing tools
- 3 `mica://` resources, with templates
- `--mcp` stdio relay
- Settings pane: enable, permissions, clients, activity
- Generated client-setup snippets, token reveal/copy/regenerate
- Fixed port + persisted token, so client config survives a restart
- Activity log, redacted on the way in
- All 16 locales translated

NOT BUILT

- Background / tray-resident mode (phase 5a)
- Run at startup — no `AutoStartManager` (phase 5b)
- Cold-starting the launcher from the stdio relay
- Modrinth search and mod-update tools
- Backup tools; export, update and per-pack settings tools
- Game-log tool and log resources
- MCP prompts (`prompts/list` , `prompts/get`)
- Live session list (it is a snapshot with a Refresh button)
- "Back up first, then continue" on destructive prompts
- Server start from TUI or headless server mode

READ THIS AS

Chapters 2–10 describe what is implemented today, verified against the source. Chapter 11 lists what is missing in detail, including the parts of the original plan that were deliberately dropped rather than merely deferred.

2 Setting it up

Two toggles in Settings and one snippet pasted into your client. The launcher generates the snippet for you, already carrying the right path, port and token, because the interesting part of it is install-specific.

2.1 Prerequisites

- **The launcher must be open.** The server lives inside the launcher process; there is no background mode yet (see §11.2). Close the launcher and the server stops, the endpoint file is deleted, and the port is released.
- **Client (GUI) mode.** `McpBootstrap.startIfEnabled()` is called from the GUI branch of `LauncherSession.run`. A TUI or headless server run does not start it — and would fail closed on any tool needing consent, because there would be no window to ask in.
- **An MCP client that speaks Streamable HTTP**, or one that can spawn a subprocess (then see §2.5).

2.2 Turning the server on

Open **Settings → Security** and scroll to **MCP Server**.

CONTROL	DEFAULT	EFFECT
Enable MCP server	OFF	Binds the loopback listener, generates the bearer token on first use, writes <code>mcp-endpoint.json</code> , and registers the read-only tools and resources. Toggling it restarts the server immediately — no launcher restart needed.
Auto-approve read-only tools	ON	Lets tools that only read state run without a prompt. Turn it off to be asked for every single call, including <code>list_modpacks</code> .
Allow tools that change things	OFF	The second opt-in — see §2.3.

The status line under the toggles reads *Running on 127.0.0.1:47823* with a count of connected clients, or *Not running*. It never shows the token.

2.3 The second opt-in

Enabling the server on its own gives you a **strictly read-only** server. The eight state-changing tools are registered only when **Allow tools that change things** is also on.

WHY TWO SWITCHES

The step from "a model can read my launcher" to "a model can install and launch things on my machine" is worth making deliberately rather than as a side effect of enabling a feature. With the second switch off, a state-changing tool is not denied — it *does not exist*. A client asking for `uninstall_modpack` gets `-32602 Unknown tool`, and `tools/list` never advertises it.

Each of those tools still asks before it runs, every time, unless you give it a standing policy in the tool-permissions list (§7.3) or grant it for the session at the prompt.

2.4 Connecting a client over HTTP

HTTP is the primary form. The launcher hosts a loopback server on a fixed port with a persisted token, so a client is configured once and stays configured across launches. In **Settings → Security → Connecting a client**, pick your client from the dropdown and press **Copy**; the snippet already contains the real port and the real token.

CLAUDE CODE

```
claude mcp add --transport http mica-launcher http://127.0.0.1:47823/mcp \
  --header 'Authorization: Bearer <token>'
```

CURSOR — ~/.CURSOR/MCP.JSON

```
{
  "mcpServers": {
    "mica-launcher": {
      "url": "http://127.0.0.1:47823/mcp",
      "headers": {
        "Authorization": "Bearer <token>"
      }
    }
  }
}
```

CODEX — ~/.CODEX/CONFIG.TOML

```
[mcp_servers.mica-launcher]
url = "http://127.0.0.1:47823/mcp"

[mcp_servers.mica-launcher.http_headers]
Authorization = "Bearer <token>"
```

ANYTHING ELSE

Transport is Streamable HTTP: POST JSON-RPC to `http://127.0.0.1:<port>/mcp` with `Content-Type: application/json` and `Authorization: Bearer <token>`. Carry the `Mcp-Session-Id` returned by the handshake on every subsequent request.

CLIENT CONFIG FORMATS DRIFT

The snippets are generated from what these clients accepted at the time of writing. If a client rejects one, check its own documentation — the URL and the `Authorization` header are the parts that matter, and every client has some way to express them.

2.5 The stdio relay fallback

Some clients can only spawn a subprocess and talk newline-delimited JSON-RPC over its stdin and stdout. For those, register the launcher with `--mcp` :

```
# native install
'/Applications/Mica Minecraft Launcher.app/Contents/MacOS/Mica Minecraft Launcher' --mcp

# fat JAR
java -jar /path/to/MicaMinecraftLauncher-jar-with-dependencies.jar --mcp
```

That process is **not** a second launcher. It is a thin relay (`StdioProxy`) that reads `mcp-endpoint.json` , forwards each line to the running launcher's HTTP endpoint, and writes the reply back. It has two properties worth knowing:

- **No token in your client's config.** The relay reads port and token from the endpoint file itself.
- **It interprets nothing.** Admission checks and approval decisions happen in the launcher exactly as they would for a direct HTTP client, so a compromised relay gains nothing a local process could not already attempt against the port.

The relay is dispatched from `LauncherCore.main()` *before* the single-instance lock is taken, so it works alongside an already-open launcher instead of being turned away by it. `stdout` carries protocol only — every diagnostic goes to `stderr` , and the relay never touches the launcher's logger.

NO COLD START

If no launcher is running with the server enabled, the relay prints an explanation to `stderr` and exits with code **2**. Cold-starting the launcher from the relay was planned and is not implemented; open the launcher first.

2.6 Ports and the endpoint file

THING	VALUE
Release port	47823 (<code>LauncherConstants.MCP_SERVER_PORT</code>)
Dev-build port	47824, so a dev and a release launcher never collide
Override	config key <code>mcpPort</code> ; there is no UI for it
Path	<code>/mcp</code> — any other path is not served
Endpoint file	<code><config folder>/mcp-endpoint.json</code> , mode <code>0600</code> (ACL on Windows)
Contents	<code>{ "port", "token", "pid", "launcherVersion" }</code>
Lifecycle	Written on start, deleted on shutdown. A file whose recorded pid is dead reads as absent

If the configured port is already taken, the server falls back to an OS-assigned port rather than refusing to start. The Settings pane and the endpoint file both report the port actually bound, so an HTTP client configured with the fixed port will need updating in that case — the stdio relay picks it up automatically.

2.7 The bearer token

256 bits from `SecureRandom`, hex-encoded to 64 characters, stored in launcher config and reused across launches.

WHY IT PERSISTS

A client's configuration names a URL and a token and is edited by hand. A token that rotated every launch would mean re-editing every client config on every start — which in practice means the feature goes unused or gets worked around. A stable token is the trade every local HTTP service makes, with the same mitigations: loopback only, owner-only storage, and a **Regenerate** button that invalidates every client still holding the old one and restarts the server on the spot.

In Settings the token is masked (`3f9a21c8... (64 characters)`) with **Show**, **Copy token** and **Regenerate** beside it. A stored value that is not a well-formed 64-character hex string — blank, truncated, hand-edited — is replaced rather than used, because serving a malformed token would leave the server running with a credential no client could ever match, which looks like a hang rather than a misconfiguration.

2.8 Smoke-testing with curl

Read the port and token out of the endpoint file, then handshake:

```
CFG=~/.MicaMinecraftLauncher/config/mcp-endpoint.json
PORT=$(python3 -c "import json;print(json.load(open('$CFG'))['port'])")
TOK=$(python3 -c "import json;print(json.load(open('$CFG'))['token'])")

curl -si -X POST "http://127.0.0.1:$PORT/mcp" \
  -H "Authorization: Bearer $TOK" \
  -H 'Content-Type: application/json' \
  -d '{"jsonrpc":"2.0","id":1,"method":"initialize",
    "params":{"protocolVersion":"2025-06-18","capabilities":{},
              "clientInfo":{"name":"curl","version":"1"}}}'
```

The response carries `Mcp-Session-Id`. Send it back on every following request; without it the launcher opens a new, uninitialized session and refuses everything but `ping` with `-32600 Session is not initialized`. Appendix B has the rest of the calls.

2.9 Troubleshooting

SYMPTOM	CAUSE AND FIX
401 with no body	Missing or wrong bearer token. Re-copy it from Settings; if you pressed Regenerate , every client config needs updating.
403	The peer is not loopback, or the request carried an <code>Origin</code> that is not itself loopback — including <code>Origin: null</code> . Browser pages are refused by design; see §4.2.
405	Not a POST. The endpoint answers POST only; there is no GET/SSE stream.

SYMPTOM	CAUSE AND FIX
413	Body over 1 MB. A JSON-RPC tool call is small; this cap exists so a local process cannot exhaust launcher memory.
415	Content-Type is not application/json.
Connection refused	The launcher is closed, the server is off, or it bound a fallback port. Check the Settings status line, then the endpoint file.
-32600 Session is not initialized	A call arrived before initialize, or the Mcp-Session-Id header is not being carried.
-32602 Unknown tool: uninstall_modpack	Allow tools that change things is off. That is the second opt-in working as intended.
-32003 Tool call was not approved	Policy said DISABLED, you denied the prompt, the prompt timed out after 120 s, or there was no window to show it in.
Tool call hangs for two minutes	A consent dialog is waiting behind another window. Every call is serialized, so one unanswered prompt stalls the rest until it times out.
Relay exits with code 2	No running launcher with the server enabled. Start the launcher; there is no cold start.
Client says "server has no tools"	The handshake succeeded against a server whose master switch was on but which has since stopped, or the client cached an old tools/list. Reconnect.

3 Architecture

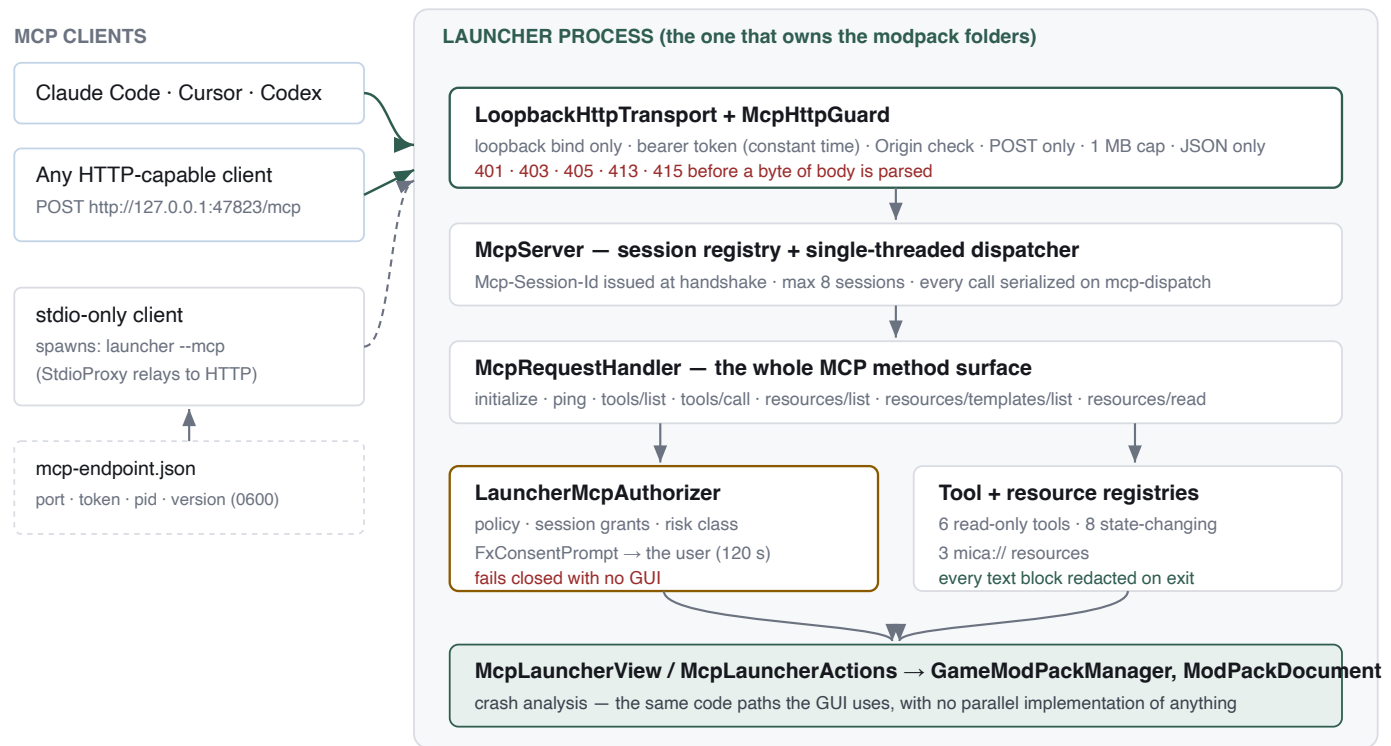


Figure 1. Clients reach the launcher over loopback HTTP, or through the `--mcp` relay which reads the endpoint file and speaks HTTP on their behalf. Everything below the admission layer runs inside the launcher process that owns the modpack folders.

3.1 Why the server lives in the launcher

The launcher enforces single-instance because concurrent processes mutating the same modpack folders is unsafe: `SynchronizedFileManager` serializes file access *in-process* and knows nothing about a second JVM. A standalone MCP server that touched packs directly was therefore never an option. Every deployment shape ends at the same place — tools execute inside the one launcher process — and the stdio relay exists precisely because MCP clients want to spawn a subprocess and the launcher cannot be one.

3.2 Package map

PACKAGE / CLASS	RESPONSIBILITY
mcp.McpBootstrap	The only lifecycle entry point. Owns the single server instance, the grant store, the activity log and the policy store; decides whether to start at all.
mcp.McpServer	Binds the transport, publishes the endpoint file, owns the session registry and the single-threaded dispatcher.
mcp.McpRequestHandler	The whole MCP method surface, with no transport or JavaFX dependency — which is what lets the protocol be tested headlessly.
mcp.McpAccessToken	Generate, validate, persist and mask the bearer token.

PACKAGE / CLASS	RESPONSIBILITY
mcp.McpEndpointFile	Write, read, stale-check and delete mcp-endpoint.json.
mcp.McpClientSetup	Generates the per-client configuration snippets shown in Settings.
mcp.McpActivityLog	Bounded, in-memory, redacted-on-write record of what clients did.
mcp.FxConsentPrompt	The consent dialog bridge; headless-safe and deadline-bounded.
mcp.protocol.*	JsonRpcCodec (parse, frame, build errors), JsonRpcMessage, McpMethods, McpErrors.
mcp.transport.*	LoopbackHttpTransport (the listener), McpHttpGuard (pure admission checks), StdioProxy (the relay).
mcp.session.*	McpSession (identity, counters, init state), McpSessionRegistry (live sessions, cap of 8).
mcp.tools.*	McpTool, McpToolRegistry, McpToolResult (the redaction boundary), ReadOnlyTools, MutatingTools, and the McpLauncherView / McpLauncherActions seams with their live implementations.
mcp.resources.*	McpResource, McpResourceRegistry, McpResourceUri (the mica:// segment gate), LauncherResources.
mcp.approval.*	McpRiskClass, McpApprovalPolicy, McpApprovalDecision, McpApprovalEngine (pure), LauncherMcpAuthorizer (production wiring), McpToolPolicyStore, McpGrantStore.

The `McpLauncherView` / `McpLauncherActions` split is load-bearing rather than stylistic. Tools never touch launcher internals directly; they read through a view whose records have nowhere to put a token, a UUID or a filesystem path, and they act through an interface with a fixed, reviewed set of operations. That is what makes §4.6 a structural property instead of a rule someone has to remember.

3.3 The life of a tool call

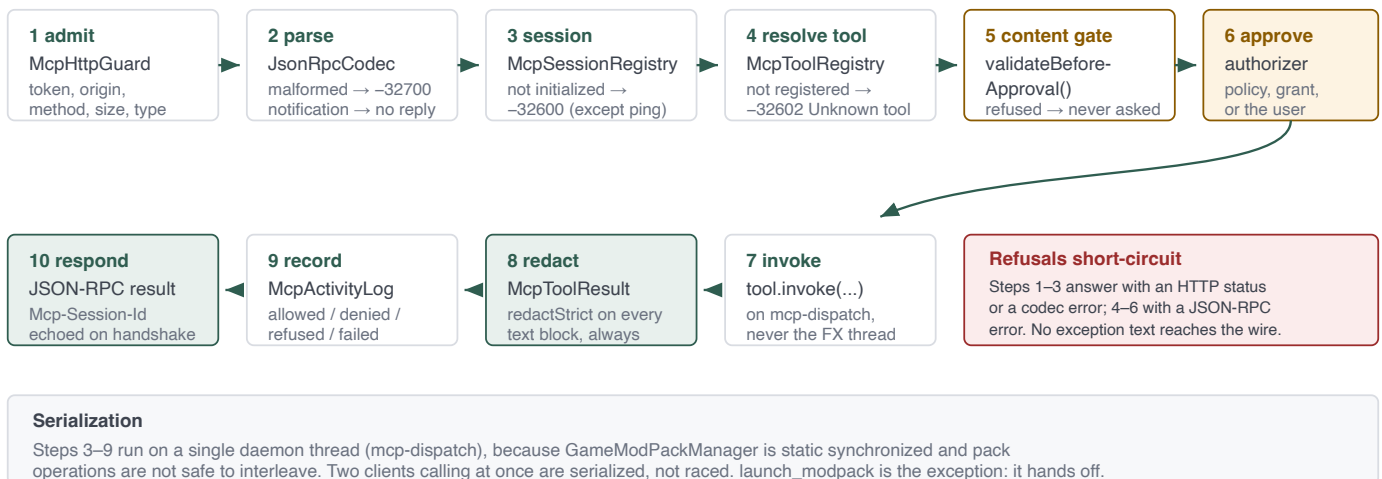


Figure 2. A `tools/call` from arrival to reply. Refusals short-circuit: the earlier the stop, the less the caller learns.

1. **Admission.** `McpHttpGuard.admit()` checks peer, origin, token, method, size and media type, in that order, before a byte of body is parsed.
2. **Parse.** `JsonRpcCodec.parse()` turns the body into a message or an error. A notification produces no reply, ever — including an unknown one, which is explicitly not an error in JSON-RPC.
3. **Session.** The `Mcp-Session-Id` header resolves to a live session, or a new one is created (subject to the cap of 8). Everything but `ping` and `initialize` requires an initialized session, because the client's self-reported name is half of every session-grant key.
4. **Tool resolution.** An unregistered name is `-32602 Unknown tool`. With state changes off, the mutating tools are genuinely not registered.
5. **Content gate.** `tool.validateBeforeApproval()` runs *before* consent: a request the launcher would refuse anyway must never reach the user as a question. Prompting for something that was never going to run is how a consent dialog becomes noise to click through.
6. **Approval.** `LauncherMcpAuthorizer` resolves policy and, when required, asks (chapter 4).
7. **Invocation** on the dispatch thread — never the FX thread.
8. **Redaction.** `McpToolResult.toJson()` passes every text block through `SensitiveDataRedactor.redactStrict`.
9. **Record.** The activity log gets the client, the tool, the decision and a short redacted detail.
10. **Reply.** A JSON-RPC result or error; the session id is echoed on the handshake only.

3.4 Protocol surface

Protocol revision `2025-06-18`. The server advertises `tools.listChanged = false` and `resources.subscribe = false`, `resources.listChanged = false` — it does not push change notifications, so a client should re-list rather than wait to be told.

METHOD	BEHAVIOUR
<code>initialize</code>	Records <code>clientInfo.name</code> and <code>.version</code> on the session, returns protocol version, capabilities and <code>serverInfo</code> (<i>Mica Minecraft Launcher</i> + version). The response carries <code>Mcp-Session-Id</code> .
<code>notifications/initialized</code>	Logged; never answered.
<code>ping</code>	Empty result. The one method allowed before the handshake.
<code>tools/list</code>	Name, title, description and JSON input schema for every registered tool.
<code>tools/call</code>	The full path in §3.3.
<code>resources/list</code>	Concrete URIs, with per-pack instances enumerated from the live pack list.
<code>resources/templates/list</code>	The <code>{friendlyName}</code> templates.
<code>resources/read</code>	Returns one text content block, redacted.
anything else	<code>-32601 Unknown method</code> .

`prompts/list` and `prompts/get` have constants in `McpMethods` but no implementation; they answer `-32601`.

ERROR CODES

CODE	NAME	WHEN
-32700	Parse error	Body is not JSON.
-32600	Invalid request	Not a valid JSON-RPC message; or a call before the handshake.
-32601	Method not found	Unknown method.
-32602	Invalid params	Missing/unknown tool name, missing argument, or a content gate refusal.
-32603	Internal error	The tool threw, a resource read threw, the session cap was hit, or dispatch failed. Never carries the exception's own message.
-32002	Resource not found	No resource matches the URI — including one rejected by the segment gate.
-32003	Request denied	The approval gate said no.

TOOL FAILURE VS PROTOCOL FAILURE

"No crash report for this pack" is not a protocol error. Tools that could not do what was asked return a normal result with `isError: true` and a message the model can act on, so it corrects itself rather than treating the server as broken. JSON-RPC errors are reserved for the protocol going wrong or the request being refused.

3.5 Sessions

- Created on first contact; identified by the `Mcp-Session-Id` header.
- Hold the client's name and version, connection time, tool-call count and last activity — what the Settings *Connected clients* list shows.
- Capped at **8**. At the cap a new session is refused rather than an existing one evicted — evicting would drop someone else's handshake state mid-conversation.
- Cleared when the server stops. Session grants are cleared with them.

3.6 Threading

The transport's accept loop runs on daemon threads. Every parsed message is dispatched through a **single-threaded executor** named `mcp-dispatch`, because `GameModPackManager` is `static synchronized` and several pack operations are not safe to interleave. Two consequences follow:

- **A consent dialog blocks the queue.** Consent marshals to the FX thread and blocks dispatch until answered, denied or timed out at 120 s.
- **Launch_modpack hands off.** `LauncherCore.play()` blocks while the game downloads and prepares, so it is spawned onto its own task and the tool reports that the launch *started* rather than holding dispatch.

4 Security model

The threat to keep in front of you: `launch_modpack` runs third-party mod code under your live Microsoft session; `install_modpack` and `add_mod_to_modpack` download from model-chosen URLs; `uninstall_modpack` deletes worlds. An MCP server hands those to whatever the model decides to do. Every design choice below follows from that.

4.1 The layered gates

#	GATE	WHAT IT STOPS
1	Master switch (<code>mcpServerEnabled</code> , off)	Everything. No port binds, no token is generated, no file is written.
2	State-change switch (<code>mcpAllowStateChanges</code> , off)	All eight mutating tools — they are not registered, so they do not exist on the wire.
3	Transport admission	Non-loopback peers, browser origins, unauthenticated callers, wrong method, oversized or non-JSON bodies (§4.2).
4	Handshake ordering	Any call from a client that has not said who it is.
5	Content gates	Install URLs the launcher already refuses, path traversal in a mod's install path, editing a manifest the launcher does not own, uninstalling or launching while a game runs (§4.7).
6	Approval engine	Anything the user's policy, grants and risk-class defaults do not permit (§4.3).
7	Consent dialog	Whatever the user says no to, plus anything nobody answers within 120 s (§4.4).
8	Redaction	Credentials in any outbound text, unconditionally (§4.6).
9	Activity log	Nothing — but it is how you find out afterwards what was allowed (§8).

4.2 Transport admission

The listener binds loopback only and refuses to serve a non-loopback peer. Binding is not the whole defence, though: **the realistic attacker is a web page the user already has open**. A page can issue cross-origin requests to `127.0.0.1`, and DNS rebinding lets it do so under a hostname it controls. `McpHttpGuard` therefore applies a fixed, ordered table — every check a pure function of request metadata, which is what allows the whole table to be tested without opening a socket.

ORDER	CHECK	FAILURE	NOTES
1	Peer address is loopback	403	A peer that cannot be resolved is treated as not loopback.

ORDER	CHECK	FAILURE	NOTES
2	Origin absent, or loopback	403	Absent is normal for non-browser clients and allowed. <code>null</code> — a sandboxed iframe or file: page — is refused. Bracketed IPv6 (<code>[::1]</code>) is handled correctly.
3	Bearer token matches	401	<code>MessageDigest.isEqual</code> , constant-time in the content, so a local attacker cannot recover the token byte by byte from response timing.
4	Method is POST	405	
5	Body $\leq$ 1 MB	413	Enforced on the declared length and again while reading.
6	Content-Type: application/json	415	Parameters such as <code>; charset=utf-8</code> are ignored.

The ordering is deliberate. Peer and origin come first because they are absolute — a request from off-loopback or from a foreign browser origin is refused regardless of whether it happens to carry a valid token. Authentication comes next, so nothing beyond it can be used to probe the server without a token. Failures return the status and nothing else.

4.3 Approval resolution

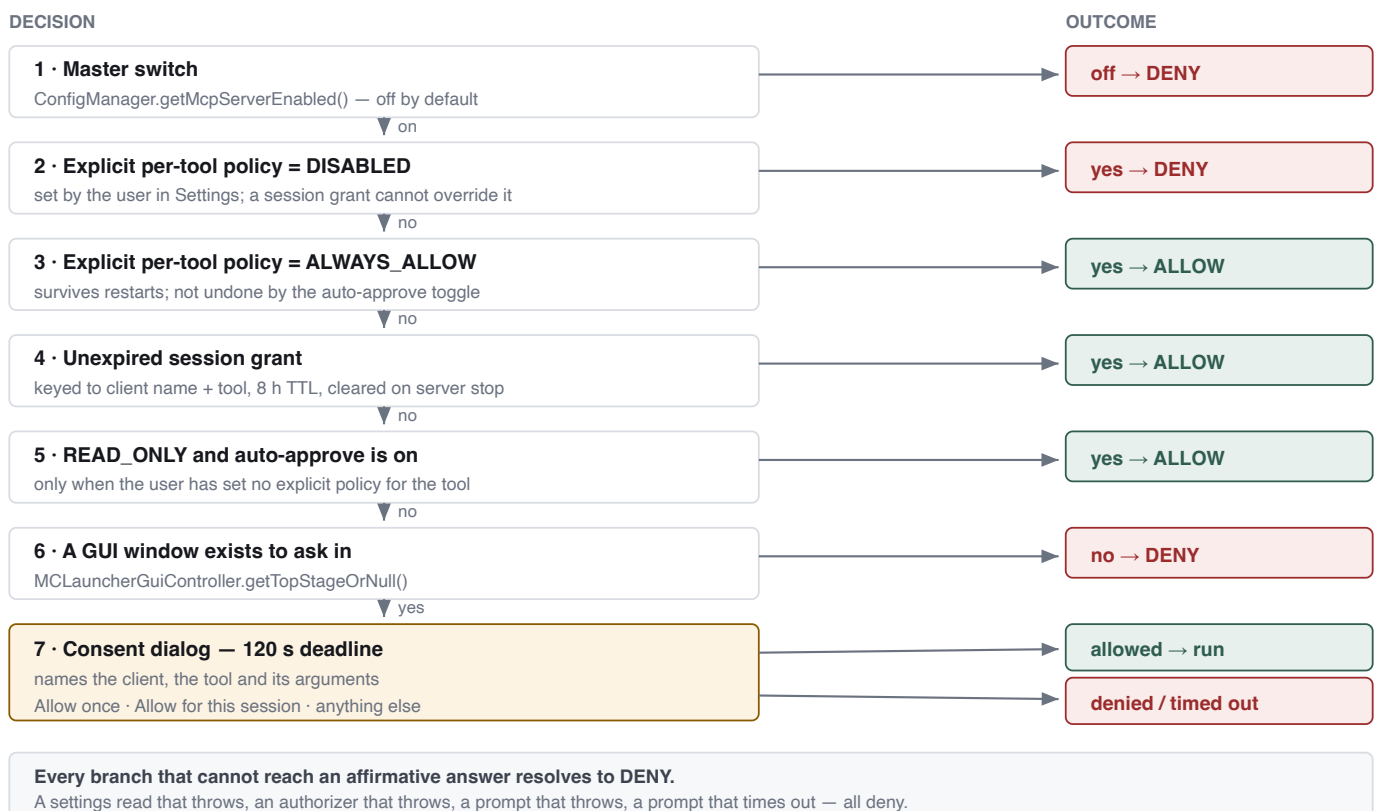


Figure 3. `McpApprovalEngine.resolve()` is a pure function of its inputs — no config reads, no clock, no UI — so the entire decision table is exercised by unit tests.

Four rules inside that order are worth stating explicitly, because each was a deliberate choice rather than a consequence:

- **A session grant cannot override `DISABLED`**. A user who turns a tool off has overruled consent they gave earlier.
- **Auto-approve gates the class default only.** Turning it off does not undo a per-tool `ALWAYS_ALLOW` the user chose deliberately: the more specific instruction wins.
- **No GUI means deny.** A launcher with no window to ask in never silently self-approves.
- **Every failure denies.** A settings read that throws, an authorizer that throws, a prompt that throws, a prompt that times out — all resolve to no.

RISK CLASSES AND THEIR DEFAULTS

CLASS	DEFAULT	MEANING
<code>READ_ONLY</code>	<code>ALWAYS_ALLOW*</code>	No state change. *Subject to the <i>Auto-approve read-only tools</i> toggle.
<code>MUTATING</code>	<code>ASK</code>	Changes launcher or pack state recoverably — install, create, fork, add or remove a mod.
<code>DESTRUCTIVE</code>	<code>ASK</code>	Deletes or overwrites user data — today, <code>uninstall_modpack</code> .
<code>EXECUTE</code>	<code>ASK</code>	Runs third-party code or spawns processes — launch and stop.

4.4 The consent dialog

Three outcomes: **Allow once**, **Allow for this session**, and deny — which is what dismissing, cancelling or pressing Escape produces, so the safe answer is the one that requires no decision.

The dialog names the **client**, the **tool** and the **arguments**, bounded: values truncated at 120 characters, at most 6 arguments rendered, structured values shown by type. An unbounded summary would push the meaningful part of the request off the bottom of the dialog.

Destructive calls add a sentence naming what would be lost. `uninstall_modpack` computes the pack's footprint first, so the question is *"this permanently deletes 4.2 GB including 3 saved worlds"* rather than *"delete this pack?"* — worlds called out separately from bytes because they are the part that is genuinely unrecoverable.

TWO DELIBERATE ABSENCES

"Always allow, permanently" is not a button. A grant that survives restarts is a considered decision, not one to make while a model waits; it lives in Settings next to the tool's risk class, where it can be reviewed and revoked.

"Back up first, then continue" is not implemented. The destructive prompt names what would be lost but does not yet offer to save it. Take a backup yourself before approving an uninstall you are unsure about.

The wait is bounded at **120 seconds**; on timeout the call is denied. The dialog itself stays on screen — the underlying helper offers no way to dismiss it — but the answer is already decided, which is the fail-closed direction.

4.5 Policies and session grants

	PER-TOOL POLICY	SESSION GRANT
Set from	Settings → tool permissions list	The consent dialog's <i>Allow for this session</i>
Values	Default · Always allow · Always ask · Disabled	Allow, implicitly
Keyed by	Tool name	Client name + tool name
Lifetime	Persisted in launcher config; survives restarts	8 hours, and cleared whenever the server stops
Precedence	Beats grants in both directions	Only applies where no explicit policy decides the call

Grants are keyed to the client's *name* rather than to the transport connection because MCP clients reconnect frequently; a connection-scoped grant would re-prompt constantly and train the user to click through without reading. The 8-hour TTL is what stops "allow for this session" from quietly becoming permanent.

An unreadable or corrupt stored policy resolves to *unset*, never to a policy — so a damaged config degrades to "ask", never to standing permission.

4.6 The credential invariant

ABSOLUTE

No tool, resource, error message or activity-log entry may ever return a credential. This is not a policy the user can relax and not something an `ALWAYS_ALLOW` can override — it is enforced in the serialization layer, below the approval engine.

Never returned: Minecraft/Microsoft access tokens, client tokens, refresh tokens; the account UUID; anything from the AES-256-GCM token cache or its machine-derived key; the MCP bearer token, the single-instance IPC token, or the endpoint file's contents; raw `${auth_access_token}` / `${auth_session}` expansions. **Permitted:** the account *username*, in `get_launcher_status` only — useful context, not a credential.

Three mechanisms enforce it, in order of how much they are relied upon:

- Nothing dangerous is reachable.** Tools read through `McpLauncherView`, whose records have no field for a token, a UUID or a path. The primary control is the shape of the seam, not a filter.
- Every text block is redacted on the way out.** `McpToolResult.toJson()` runs `SensitiveDataRedactor.redactStrict` over each block — access-token and client-token flags in both `--flag value` and `--flag=value` forms, the legacy `token:<token>:<uuid>` session string, dashed and undashed UUIDs. Resource text goes through the same path. A tool *cannot* return unredacted text through this type, including one written later by someone who has not read this document.

3. **The activity log redacts on the way in**, so a token that reached an argument summary is never held in memory by it at all — a Settings page is one screenshot away from a support thread.

The dangerous surface is logs and crash reports, which are large, attacker-influenced, and historically carried the live access token on the game's command line. SHA-1 hashes survive redaction deliberately; they are how mods are identified.

4.7 Containment rules

RULE	ENFORCED WHERE
Resource URI segments	<code>McpResourceUri.isSafeSegment</code> : no <code>/</code> , no <code>\</code> , no NUL, no control characters, not <code>.</code> or <code>...</code> . Percent-decoding that fails leaves the raw text in place, so a deliberately malformed encoding cannot slip a separator through unchecked. Listings skip packs whose names are not safe segments, so a listing never advertises a URI the reader would refuse.
Mod install paths	<code>MutatingTools.rejectUnsafeLocalPath</code> : no absolute paths, no drive letters, no <code>..</code> segment, no control characters. Checked <i>before</i> consent — a traversal attempt is not something to ask the user to adjudicate.
Install and mod URLs	<code>LauncherUriHandler.classifyInstallUrl()</code> , the launcher's existing install-source gate. A REJECT verdict refuses before any prompt; REQUIRE_CONFIRMATION deliberately does not — an untrusted but well-formed host is exactly what the consent dialog is for.
Editable manifests	Only manifests this launcher authored — resolved inside <code><config>/mcp-manifests</code> after normalisation — may be edited in place. A pack installed from a remote manifest, or an imported <code>.mmcjson</code> , is refused with a message pointing at <code>fork_modpack</code> .
Live-game guards	<code>uninstall_modpack</code> refuses while that pack's game runs; <code>launch_modpack</code> refuses a pack that is already running, an account already playing another pack, and a launch with nobody signed in. Several games may run at once otherwise.

4.8 Threat model and residual risk

RISK	POSITION
Any same-UID process can read the endpoint file	Accepted, and identical to the trust boundary the launcher's existing single-instance IPC channel already documents. The file is <code>0600</code> ; a process running as you can read it, and a process running as you could also read your session token from the launcher's own config.
A malicious or confused model	The core case the consent model exists for. Mutating tools are off by default and ask every time; the prompt names the client, the tool and the arguments; the activity log records what happened.
Consent fatigue	Partially addressed: read-only auto-approval, session grants with a TTL, and batch-shaped tools that return lists rather than one item at a time. Per-client rate limiting and a <i>deny all from this client</i> action are not implemented.

RISK	POSITION
A web page in the user's browser	Blocked by the origin rule plus a token the page cannot read.
Token theft from a client's config file	Real, and the reason Regenerate exists. The stdio relay avoids it entirely by never putting a token in client config.
A model choosing a hostile install URL	The launcher's existing install-source classification runs first; anything it does not outright reject is shown to the user in the prompt, naming the host.
Two clients issuing conflicting mutations	Serialized by the single-threaded dispatcher. Destructive tools re-check state inside their own validation before acting.
An unattended launcher	Prompts deny after 120 s rather than waiting forever. The stale dialog left on screen is cosmetic, but it is a known rough edge.

5 Tool reference

Fifteen tools: seven that only read, eight that change things. The seven are registered whenever the server runs; the eight appear only when *Allow tools that change things* is on.

5.1 Conventions

- **Packs are addressed by friendly name** — the display name, exactly as `list_modpacks` reports it. Names routinely contain spaces, colons and version numbers (`Fabric 1.21.5: 0.16.10`); pass them verbatim.
- **Arguments are strings** and are trimmed. A missing or non-primitive value reads as empty and produces a clear error naming what was required.
- **"Not found" is a tool error, not a protocol error.** Results carry `isError: true` and a message naming `list_modpacks` so the model can recover on its own instead of guessing again.
- **Structured output is a pretty-printed JSON text block**, which is how MCP carries structured tool output.
- **Every text block is redacted** on the way out (§4.6).
- **Content gates run before consent**, so an invalid call never becomes a dialog.

5.2 Read-only tools

All six are `READ_ONLY` and run without a prompt while *Auto-approve read-only tools* is on, which it is by default.

`list_modpacks`

`READ_ONLY` · always registered

PURPOSE Every pack the launcher knows about, installed and available. The starting point for everything else — it is where friendly names come from.

ARGUMENTS None.

RETURNS `modpacks[]` and `count`. Each entry: `friendlyName`, `version` (installed), `latestVersion` (published), `updateAvailable`, `modLoader`, `installed`, `unstable`.

TWO VERSIONS, ON PURPOSE

A pack's friendly name often embeds the version its author publishes, so a pack called *Alto: 26.9.3* can correctly report `version 26.6.12` — the one installed. Both numbers are right and neither can be inferred from the other, which is why both are reported. This was found by running the server against real packs, not by a test.

get_modpack_info

READ_ONLY · always registered

PURPOSE	The same fields as a <code>list_modpacks</code> entry, for one pack.
ARGUMENTS	<code>friendlyName</code> (<i>required</i>) — the pack's display name.
RETURNS	<code>friendlyName</code> , <code>version</code> , <code>latestVersion</code> , <code>updateAvailable</code> , <code>modLoader</code> , <code>installed</code> , <code>unstable</code> .
ERRORS	Missing argument; no such pack (the message names <code>list_modpacks</code>).

get_modpack_manifest

READ_ONLY · always registered

PURPOSE	The raw manifest JSON: mod, config, resource-pack and shader lists, mod-loader URLs, metadata.
ARGUMENTS	<code>friendlyName</code> (<i>required</i>).
RETURNS	The manifest as one text block.
ERRORS	No manifest available — the pack may not exist, or its manifest may be unreachable.
SEE ALSO	The same content as a re-readable resource: <code>mica://modpack/{friendlyName}/manifest</code> .

get_crash_report

READ_ONLY · always registered

PURPOSE	The most recent crash report for a pack, as raw text.
ARGUMENTS	<code>friendlyName</code> (<i>required</i>).
RETURNS	The report text, credentials stripped.
ERRORS	No crash report on record.
PREFER	<code>diagnose_launch_failure</code> — same report plus the launcher's own analysis.

diagnose_launch_failure

READ_ONLY · always registered

PURPOSE	The highest-value tool here. Turns "why did my game crash" into a question a model can answer, without anyone pasting logs around.
ARGUMENTS	<code>friendlyName</code> (<i>required</i>).
RETURNS	<code>friendlyName</code> , <code>version</code> , <code>modLoader</code> , <code>unstable</code> , <code>crashReportAvailable</code> , and when a report exists: <code>category</code> , <code>title</code> , <code>summary</code> , <code>suggestions[]</code> , <code>crashReport</code> .
NO CRASH?	Returns a success with <code>crashReportAvailable: false</code> and a note. Reporting "this pack has not crashed" as a failure would push the model to retry pointlessly.

```
{ "name": "diagnose_launch_failure",
  "arguments": { "friendlyName": "All the Mods 9" } }

→ { "friendlyName": "All the Mods 9", "version": "0.2.60",
    "modLoader": "forge", "unstable": false,
    "crashReportAvailable": true,
    "category": "MISSING_DEPENDENCY",
    "title": "A mod is missing a dependency",
    "summary": "...", "suggestions": [ "...", "..." ],
    "crashReport": "----- Minecraft Crash Report -----\\n..." }
```

get_launcher_status

READ_ONLY · always registered

PURPOSE Launcher-level context: is anyone signed in, which build is this, how many packs are installed.

ARGUMENTS None.

RETURNS `launcherVersion`, `signedIn`, `username`, `installedModpackCount`.

USERNAME ONLY

The account UUID and every token are forbidden here by §4.6, and `McpLauncherView` deliberately offers no way to reach them. Signed out, `username` is empty.

list_running_games

READ_ONLY · always registered

PURPOSE What is launching or running now. Several games can run side by side, one per modpack and one per account.

ARGUMENTS None.

RETURNS `games`: each with `friendlyName`, `account` (username), `phase` (`preparing` or `running`) and `uptimeSeconds`.

USERNAME ONLY

As with `get_launcher_status`, the playing account appears by username alone; its UUID and tokens are unreachable.

5.3 State-changing tools

SECOND OPT-IN REQUIRED

These eight are registered only when **Settings → Security → Allow tools that change things** is on. With it off they are not denied — they are absent from `tools/list` and answer `-32602 Unknown tool`. Each still asks before every call unless you have given it a standing policy.

install_modpack

MUTATING · asks by default

PURPOSE	Installs a modpack from a manifest URL, downloading mods and configuration from third-party hosts.
ARGUMENTS	<code>url</code> (<i>required</i>) — the https URL of the manifest.
BEFORE ASKING	The URL goes through <code>LauncherUriHandler.classifyInstallUrl()</code> . A REJECT verdict — non-https, malformed, control characters — is refused outright and never becomes a prompt. An untrusted-but-well-formed host is <i>not</i> rejected here: that is exactly what the consent dialog is for, and refusing it would make the tool useless for any pack not on the allowlist.
RETURNS	A sentence naming what was installed, or a failure message. The underlying install runs synchronously on the dispatch thread, so the reply means the install finished.

create_modpack

MUTATING · asks by default

PURPOSE	Creates a new, empty pack and installs it locally, ready to have mods added.
ARGUMENTS	<code>name</code> (<i>required</i>); <code>modLoader</code> (<code>forge</code> · <code>neoforge</code> · <code>fabric</code> , default <code>forge</code>); <code>modLoaderUrl</code> .
BEFORE ASKING	Refuses an empty name or a name already in use.
WRITES	A manifest into <code><config>/mcp-manifests</code> , then installs from that local URL — the same convention the <code>.mmcjson</code> and <code>.mrpack</code> importers already use. Packs created this way are the ones the edit tools may later modify.

fork_modpack

MUTATING · asks by default

PURPOSE	Copies an installed pack under a new name and installs the copy, so it can be edited without touching the original. Keeps every mod, config and setting, with the minor version bumped.
ARGUMENTS	<code>friendlyName</code> (<i>required</i>), <code>newName</code> (<i>required</i>).
BEFORE ASKING	Both names required; source must exist; target name must be free.
WHY IT MATTERS	This is the supported route to editing a pack you installed from someone else's manifest — the launcher will not rewrite a manifest it does not own (§4.7).

add_mod_to_modpack

MUTATING · asks by default

PURPOSE	Adds a mod to a pack's manifest, to be downloaded on the next sync.
ARGUMENTS	<code>friendlyName</code> , <code>modName</code> , <code>remoteUrl</code> , <code>localPath</code> — all required. <code>localPath</code> is relative to the pack root, e.g. <code>mods/jei.jar</code> .
BEFORE ASKING	Pack must exist. <code>remoteUrl</code> faces the same install-source gate as a pack manifest, for the same reason: it is a model-chosen address that will be downloaded and put on the game's classpath. <code>localPath</code> must be relative, contain no <code>..</code> segment and no control characters.

EDITABILITY

Only packs whose manifest this launcher authored. Otherwise the error points at `fork_modpack`.

remove_mod_from_modpack

MUTATING · asks by default

PURPOSE	Removes a mod from a pack's manifest; the file goes on the next sync.
ARGUMENTS	<code>friendlyName</code> (<i>required</i>), <code>mod</code> (<i>required</i>) — its local path, display name, or Modrinth slug.
BEFORE ASKING	Both required; pack must exist. The same authored-manifest rule applies.

uninstall_modpack

DESTRUCTIVE · asks by default

PURPOSE	Permanently removes a pack and deletes its files, including any worlds saved inside it. This cannot be undone.
ARGUMENTS	<code>friendlyName</code> (<i>required</i>).
BEFORE ASKING	Pack must exist, and no game may be running . Deleting files out from under a live game corrupts whatever it writes next, and the user would blame the game rather than a tool call they approved minutes earlier.
PROMPT	Names the footprint: <i>"This permanently deletes 4.2 GB including 3 saved worlds. This cannot be undone."</i> Sizes may be reported as "at least" when the walk was bounded.

NO BACKUP OFFER YET

"Back up first, then continue" is specified but not implemented. The prompt tells you what you are about to lose; saving it first is still manual.

launch_modpack

EXECUTE · asks by default

PURPOSE	Starts Minecraft for a pack. It plays as the pack's usual account (the pack's own <i>Launch as</i> setting, else the default account) unless <code>account</code> names another. This runs third-party mod code.
ARGUMENTS	<code>friendlyName</code> (<i>required</i>); <code>account</code> (<i>optional</i>), the username of a signed-in account to play as for this launch.
BEFORE ASKING	Pack must exist <i>and be installed</i> ; an account must be signed in; the pack must not already be running; the account must exist, be usable, and not already be playing another pack.
RETURNS	That the launch started . The launch is spawned onto its own task so the single dispatch thread is never held while the game prepares. Progress shows in the launcher's Running Games window; <code>list_running_games</code> follows it.

stop_game

EXECUTE · asks by default

PURPOSE	Stops a running game, or cancels one still preparing. Unsaved progress since the last autosave is lost.
ARGUMENTS	<code>friendlyName</code> (<i>optional when exactly one game is running</i>).
BEFORE ASKING	Refuses when no game is running, when several are running and none is named (it lists them), and when the named pack isn't running. The consent dialog names the game it will stop.
BEHAVIOUR	Asks that game's process to exit (<code>destroy()</code>), naming the pack it stopped.

6 Resource reference

Resources cover what a client wants to *read repeatedly*, as opposed to act on. They read through the same `McpLauncherView` as the tools, so they are subject to the same bound on what they can reach, and their text leaves through the same redaction path.

6.1 URI scheme and the segment gate

All resources use the `mica://` scheme. Per-pack resources are templates over `{friendlyName}`, and they enumerate their instances from the live pack list, so a client can call `resources/list` and see one entry per pack rather than guessing names. Names are percent-encoded in the emitted URIs —

`mica://modpack/Fabric%201.21.5%3A%200.16.10/manifest` — and decoded on the way back in.

DECODE FIRST, THEN CHECK

The containment checks run on the *decoded* segment. Validating the encoded form and decoding afterwards is the classic way this kind of gate is bypassed — `%2e%2e%2f` looks harmless until it doesn't. A segment containing `/`, `\`, NUL or any control character, or equal to `.` or `..`, is refused, and a percent sequence that fails to decode is checked in its raw form rather than being let through. A listing skips any pack whose name is not a safe segment, so it never advertises a URI that the reader would then refuse.

A URI that matches nothing — including one rejected by the gate — answers `-32002 No such resource`.

6.2 The three resources

URI	MIME	CONTENT
<code>mica://packs</code>	<code>application/json</code>	The pack index: every pack, installed and available, with the same seven fields <code>list_modpacks</code> returns. Pretty-printed.
<code>mica://modpack/{friendlyName}/manifest</code>	<code>application/json</code>	The raw manifest JSON for one pack. Enumerated for every pack, installed or not.
<code>mica://modpack/{friendlyName}/crash-report</code>	<code>text/plain</code>	The most recent crash report for one pack, credentials stripped. Enumerated for <i>installed</i> packs only.

The server advertises `resources.subscribe = false` and `listChanged = false`: it does not push notifications when packs change, so a long-lived client should re-list rather than wait to be told.

7 Settings reference

Everything lives in **Settings** → **Security**, below the malware scanner and the `mmcl://` deep-link toggle, in three cards.

7.1 Server controls

CONTROL	BEHAVIOUR
Enable MCP server	Writes <code>mcpServerEnabled</code> and calls <code>McpBootstrap.refresh()</code> , which stops and restarts the server in place.
Auto-approve read-only tools	Writes <code>mcpAutoApproveReadOnly</code> . Read immediately by the authorizer — no restart needed.
Allow tools that change things	Writes <code>mcpAllowStateChanges</code> and refreshes, because the tool registry is built at start. Connected clients should re-run <code>tools/list</code> .
Status line	<i>Running on 127.0.0.1:<port></i> or <i>Not running</i> , plus the connected-client count. Never shows the token.

7.2 Connecting a client

CONTROL	BEHAVIOUR
Endpoint	<code>http://127.0.0.1:<port>/mcp</code> , showing the port actually bound.
Token + Show / Hide	Masked as <code>3f9a21c8...</code> (64 characters) until revealed.
Copy token	Puts the full token on the clipboard.
Regenerate	Confirms first, then replaces the token and restarts the server. Every client configured with the old token stops working until updated.
Client dropdown + Copy	Claude Code · Cursor · Codex · Other (raw URL) · stdio-only client. The snippet below updates with the selection and already contains the real port and token.

RUNNING FROM SOURCE

In a development run the resolved launcher path is the JVM itself, and `<jvm>/bin/java --mcp` would start a bare JVM that does nothing. The stdio snippet is replaced with a notice saying so and pointing at the HTTP options, rather than emitting a command that is accepted and then silently fails.

7.3 Tool permissions, clients, activity

The third card holds three lists and four buttons.

- **Tool permissions.** One row per tool — every tool this build can expose, whether or not the server is running and whether or not state changes are enabled, so you can pre-configure a permission before

turning the thing on. Each row offers *Default*, *Always allow*, *Always ask*, *Disabled*. *Default* means "follow the risk class": read-only follows the auto-approve toggle; everything else asks.

- **Connected clients.** Name, connection age and tool-call count per session, or *No MCP client is connected*. This is a **snapshot**, refreshed when the pane is built and when you press *Refresh* — a live list needs an observable session registry, which does not exist yet.
- **Recent activity.** See chapter 8.
- **Revoke session approvals.** Clears every "allow for this session" grant held right now. Permanent per-tool choices are unaffected. The label above it counts what is currently held.

8 Auditing what happened

Consent answers "may this happen?". The activity log answers "what happened?". Without it, a user who clicked through a prompt three hours ago has no way to find out what they agreed to, which makes every approval decision unreviewable after the moment it is made.

8.1 The activity log

PROPERTY	VALUE
Scope	In memory, bounded to the most recent 200 entries, newest shown first.
Lifetime	Survives the server being stopped and restarted within one launcher run — toggling the feature off and on does not erase the record of what happened before, which is exactly what a user goes looking for after something surprising. It does not survive a launcher restart.
Fields	Timestamp, client name, tool name, decision, short detail.
Redaction	Every field passes <code>redactStrict</code> <i>before</i> storage, so a credential that reached an argument summary is never held in memory here at all.
Clearing	<i>Clear activity</i> in Settings.

DECISION	MEANING
refused	A content gate said no before the user was ever asked — bad URL, unsafe path, game running, unknown pack.
denied	Policy said no, the user said no, the prompt timed out, or there was no GUI to ask in.
allowed	Ran and succeeded.
failed	Ran and failed. Exception text is deliberately not recorded — it carries paths and can carry command lines.

8.2 The launcher log

The in-memory log is a review surface for the session, not an audit trail. What persists:

- The launcher's own log records server start/stop with the bound port, every approval outcome (`MCP allowed <client> -> <tool> (ALLOW_ONCE)`), every pre-approval refusal with its reason, and every tool failure with its stack trace.
- Pack-mutating operations additionally reach the launcher's existing `ModPackAuditLog` , through the same code paths the GUI uses.
- No exception message and no credential ever reaches the wire; the detail stays in the log where it belongs.

9 Configuration and file locations

9.1 Config keys

All live in the launcher's `configuration.json`, written through `ConfigStore` like every other setting.

KEY	TYPE	DEFAULT	MEANING
<code>mcpServerEnabled</code>	boolean	false	The master switch.
<code>mcpAllowStateChanges</code>	boolean	false	Registers the eight state-changing tools.
<code>mcpAutoApproveReadOnly</code>	boolean	true	Read-only tools skip the prompt.
<code>mcpToolPolicies</code>	JSON object	empty	Tool name → ALWAYS_ALLOW / ASK / DISABLED. An unreadable value resolves to "unset", never to a policy.
<code>mcpPort</code>	int	47823 (dev 47824)	The loopback port. No UI; edit the file to change it.
<code>mcpToken</code>	string	empty	The persisted bearer token. Generated on first use; a malformed value is replaced.

9.2 Files on disk

PATH	PURPOSE
<code>~/.MicaMinecraftLauncher/config/</code>	The client-mode config folder. A dev build uses <code>~/.MicaMinecraftLauncherDEV/</code> .
<code>.../config/configuration.json</code>	All six keys above.
<code>.../config/mcp-endpoint.json</code>	Port, token, pid, launcher version. Owner-only; deleted on shutdown; a dead pid makes it read as absent.
<code>.../config/mcp-manifests/</code>	Manifests this launcher authored through <code>create_modpack</code> / <code>fork_modpack</code> . Only packs whose manifest resolves inside this directory may be edited in place.

The endpoint file deliberately uses the *client-mode* config path rather than the game-mode-dependent one, for the same reason `SingleInstanceLock` does: a client discovering the endpoint has a different working directory from the launcher, so a relative path would never agree between them.

9.3 Constants

CONSTANT	VALUE
<code>LauncherConstants.PROGRAM_ARG_MCP</code>	<code>--mcp</code>
<code>LauncherConstants.MCP_SERVER_PORT / _DEV</code>	47823 / 47824

CONSTANT	VALUE
McpRequestHandler.PROTOCOL_VERSION	2025-06-18
LoopbackHttpTransport.PATH	/mcp
LoopbackHttpTransport.SESSION_HEADER	Mcp-Session-Id
McpHttpGuard.MAX_BODY_BYTES	1 MB
McpSessionRegistry.MAX_SESSIONS	8
McpApprovalEngine.DEFAULT_GRANT_TTL_MS	8 hours
FxConsentPrompt.CONSENT_TIMEOUT_SECONDS	120
McpActivityLog.MAX_ENTRIES	200
McpAccessToken.TOKEN_LENGTH	64 hex characters (256 bits)
StdioProxy.EXIT_NO_LAUNCHER	exit code 2
McpClientSetup.SERVER_NAME	mica-launcher

10 Testing and verification

10.1 Unit coverage

The MCP package carries **407 test methods across 21 test classes**, in the repository's established style: plain JUnit Jupiter, hand-rolled stubs, no mocking framework, no JavaFX, no network, no vendor SDKs.

SEAM	TESTS	WHAT IS PINNED
JsonRpcCodec	34	Framing round-trip, malformed JSON, notification vs request, error construction.
MutatingTools	36	Schemas, pre-approval gates, path-traversal rejection, size formatting, destructive framing.
McpRequestHandler	33	Handshake ordering, unknown methods, denied tools, throwing tools, resource reads.
McpHttpGuard	27	The full admission table, including bracketed IPv6 origins and <code>Origin: null</code> .
McpResourceUri	27	The containment gate, with encoded spellings of every rejected character — not just the literal ones.
ReadOnlyTools	25	Every tool's success and failure shapes against a stub view.
McpClientSetup	24	Snippet generation, shell quoting of paths with spaces, TOML escaping, the unpackaged-run notice.
McpApprovalEngine	23	The decision table across master switch × policy × grant × risk class × GUI availability, asserting fail-closed.
McpServerLoopbackTest	18	A real socket: non-loopback refusal, 401 on a bad token, session issuance.
McpToolPolicyStore	18	Persistence round-trip; a corrupt value resolving to "unset".
LauncherMcpAuthorizer	17	Grant lifecycle, prompt answers, every throw-path denying.
McpToolResult	17	That no synthetic token or UUID escapes un-redacted, through results or errors.
Others	108	Endpoint file staleness, token well-formedness and masking, activity-log bounding and redaction, registries, resources, the stdio relay, the consent prompt's argument summary.

Across the whole project the suite now stands at 115 test classes. CI runs `mvn -B test` as its own job before the three-platform package matrix, so a failing MCP test blocks a PR like any other.

10.2 What a live run confirmed

Unit tests say whether the seams behave, not whether the feature works. The read path was therefore driven for real — a harness calling `McpBootstrap.startIfEnabled()` against the actual dev config and the

installed packs, over HTTP with `curl` .

EXERCISED	RESULT
Handshake, protocol version, server info	As specified; <code>Mcp-Session-Id</code> issued and carried
<code>tools/list</code> with state changes off	The six read-only tools only
<code>list_modpacks</code> , <code>get_launcher_status</code>	Real installed packs; signed-out reported with an empty username
<code>resources/list</code> and read	Correctly percent-encoded per-pack URIs that resolve back; real manifest returned
<code>diagnose_launch_failure</code> with no crash on record	Success with a note, not an error
No bearer token / wrong token	401
Origin: <code>http://evil.example</code> · Origin: <code>null</code>	403
Origin: <code>http://localhost:3000</code>	200
Content-Type: <code>text/plain</code> · GET	415 · 405
Tool call before the handshake	-32600 Session is not initialized
<code>mica://modpack/%2e%2e/manifest</code>	-32002 No such resource
<code>uninstall_modpack</code> with state changes off	-32602 Unknown tool
Endpoint file	Written <code>-rw-----</code> with port, 64-character token and pid; deleted on shutdown, port released

FOUND ONLY BY RUNNING IT

A pack's friendly name embeds the version its author publishes, so `"A/to: 26.9.3"` reported version 26.6.12 — the installed one. Both are correct and irreconcilable from the output, which is why `PackSummary` now carries `latestVersion` and `updateAvailable` as well. No unit test could have caught it: the stub returned whatever version the test asked for.

10.3 Manual checklist

These surfaces have no automated coverage. Walk them after any change:

- The Settings toggles, including that enabling and disabling restarts the server in place and the status line follows.
- The consent dialog: all three answers, the timeout, and that a denied call returns `-32003` rather than hanging.
- Every state-changing tool end to end — create, add a mod, install, launch, stop, uninstall — since these run real installs and a real game.
- A real MCP client, not just `curl` : point one at the HTTP endpoint, then at `launcher --mcp` , and ask it why a game crashed.
- Token regeneration invalidating a connected client.

11 Limitations and future work

11.1 Known limitations

LIMITATION	DETAIL
The launcher must be open	There is no background or tray-resident mode. Closing the window stops the server.
GUI mode only	<code>startIfEnabled()</code> is called from the GUI branch of <code>LauncherSession.run</code> . TUI and headless server mode never start it.
The client list is a snapshot	Refreshed on pane build and on <i>Refresh</i> . A live list needs an observable session registry.
A timed-out consent dialog stays on screen	The answer is already decided — deny — so this is cosmetic, but the stale dialog has to be dismissed by hand.
One consent prompt blocks every call	Dispatch is single-threaded by design. An unanswered prompt stalls the queue for up to two minutes.
No cold start from the relay	<code>launcher --mcp</code> exits with code 2 when no launcher is running.
Only authored manifests are editable	<code>add_mod_to_modpack</code> and <code>remove_mod_from_modpack</code> refuse a pack installed from a remote manifest. Fork it first.
No per-modpack scoping	A client granted a tool holds it for every pack; "this client may only touch pack X" does not exist.
No rate limiting	There is no per-client prompt throttle and no <i>deny all from this client</i> action.
Three English strings use "(s)"	<code>{0} tool call(s)</code> and friends do not translate cleanly; translators rendered the meaning, but the source would be better as a real plural form.

11.2 Not yet implemented

The original plan is broader than what shipped. What remains, roughly in the order it would be worth doing:

- Remaining read-only tools.** `list_modpack_mods`, `search_modrinth`, `get_modrinth_project`, `check_mod_updates`, `detect_mod_conflicts`, `get_game_log`, `list_java_runtimes`, `list_modpack_backups`, `list_server_favorites`. Each has an existing launcher class behind it; these are adapters, not new capability.
- Remaining state-changing tools.** `set_modpack_settings`, `update_modpack`, `export_modpack` (needs destination-path validation — it writes outside the launcher directory), `create_modpack_backup`, `restore_modpack_backup`, `delete_modpack_backup`.
- "Back up first, then continue"** on the destructive prompt. The prompt already names what would be lost; offering to save it is the other half.
- Remaining resources.** `.../mods`, `.../log`, and `mica:///launcher/log`.

5. **Background mode (phase 5a).** Extract tray ownership out of `NotificationManager` into a `TrayPresenceManager`, close-to-tray with `setImplicitExit(false)` under a single owner, MCP status and *Quit* in the shared shell menu, and a first-close toast so a server is never left running unknowingly. Background mode must refuse to enable where `SystemTray.isSupported()` is false — a background process with no visible way to stop it is not acceptable.
6. **Run at startup (phase 5b).** A new `AutoStartManager` with per-user backends — `HKCU\...\Run`, a `LaunchAgent` plist, `~/.config/autostart` — plus a `--background` flag distinct from `--mcp`, and config/OS reconciliation on startup where the OS wins.
7. **Cold start from the relay**, which depends on background mode existing.
8. **MCP prompts** (`diagnose_crash`, `plan_modpack`) and resource change notifications.
9. **A live session list** backed by an observable registry.

ALSO OUTSTANDING

No real MCP client has been driven against the server end to end, and the state-changing tools have not been exercised against a live launcher — the verification run in §10.2 deliberately exercised nothing that could change the machine. Both are worth doing before the feature is described as finished.

A Appendix — a worked session

One complete exchange, as a client actually performs it. Requests are `POST /mcp` with `Content-Type: application/json` and `Authorization: Bearer <token>`; every request after the first also carries `Mcp-Session-Id`.

1 · HANDSHAKE

```
→ {"jsonrpc":"2.0","id":1,"method":"initialize","params":{"protocolVersion":"2025-06-18","capabilities":{},"clientInfo":{"name":"Claude Code","version":"2.0.1"}}}

← HTTP/1.1 200 · Mcp-Session-Id: 6f1c0f0a-...
{"jsonrpc":"2.0","id":1,"result":{"protocolVersion":"2025-06-18","capabilities":{"tools":{"listChanged":false,"resources":{"subscribe":false,"listChanged":false}},"serverInfo":{"name":"Mica Minecraft Launcher","version":"3.0.0"}}}

→ {"jsonrpc":"2.0","method":"notifications/initialized"}
← HTTP/1.1 202 · (no body – a notification is never answered)
```

2 · DISCOVERY

```
→ {"jsonrpc":"2.0","id":2,"method":"tools/list"}

← {"jsonrpc":"2.0","id":2,"result":{"tools":[{"name":"list_modpacks","title":"List modpacks","description":"Lists every modpack the launcher knows about...","inputSchema":{"type":"object","properties":{}}}, {"name":"get_modpack_info","...":"..."}, {"name":"get_modpack_manifest","...":"..."}, {"name":"get_crash_report","...":"..."}, {"name":"diagnose_launch_failure","...":"..."}, {"name":"get_launcher_status","...":"..."}]}}
```

3 · A READ-ONLY CALL — NO PROMPT, BECAUSE AUTO-APPROVE IS ON

```
→ {"jsonrpc":"2.0","id":3,"method":"tools/call","params":{"name":"list_modpacks","arguments":{}}}

← {"jsonrpc":"2.0","id":3,"result":{"content":[{"type":"text","text":"\n\n\"modpacks\": [\n  {\n    \"friendlyName\": \"All the Mods 9\",\n    \"version\": \"0.2.60\",\n    \"latestVersion\": \"0.2.62\",\n    \"updateAvailable\": true,\n    \"modLoader\": \"forge\",\n    \"installed\": true,\n    \"unstable\": false\n  }\n],\n\n\"count\": 1\n}"],"isError":false}}
```

4 · A RESOURCE READ

```
→ {"jsonrpc":"2.0","id":4,"method":"resources/read","params":{"uri":"mica://modpack/All%20the%20Mods%209/manifest"}}
```

```
← {"jsonrpc":"2.0","id":4,"result":{"contents":[{"uri":"mica://modpack/All%20the%20Mods%209/manifest",
"mimeType":"application/json",
"text":{"packName\":"All the Mods 9\"," ... }]]}}
```

5 · A STATE-CHANGING CALL — THE LAUNCHER ASKS THE USER

```
→ {"jsonrpc":"2.0","id":5,"method":"tools/call","params":{"name":"launch_modpack","arguments":{"friendlyName":"All the Mods 9"}}
```

... the launcher shows: "Claude Code wants to use \"Launch modpack\"", naming the tool, its description and its arguments. The request is held – for up to 120 seconds – until the user answers.

```
← (allowed) {"jsonrpc":"2.0","id":5,"result":{"content":[{"type":"text",
"text":"Started launching \"All the Mods 9\". The launcher window shows
progress; the game takes a while to appear."}], "isError":false}}
```

```
← (denied) {"jsonrpc":"2.0","id":5,"error":{"code":-32003,
"message":"Tool call was not approved: launch_modpack"}}
```

6 · A REFUSAL THAT NEVER REACHES THE USER

```
→ {"jsonrpc":"2.0","id":6,"method":"tools/call","params":{"name":"install_modpack","arguments":{"url":"ftp://example.com/pack.json"}}
```

```
← {"jsonrpc":"2.0","id":6,"error":{"code":-32602,
"message":"That install URL was refused by the launcher's install-source
rules. Modpack URLs must be well-formed https links."}}
```

B Appendix — curl cookbook

Everything below assumes `$PORT`, `$TOK` and `$SID` are set as in §2.8. Useful for confirming the server is healthy without involving a client.

HELPERS

```
CFG=~/.MicaMinecraftLauncher/config/mcp-endpoint.json
PORT=$(python3 -c "import json;print(json.load(open('$CFG'))['port'])")
TOK=$(python3 -c "import json;print(json.load(open('$CFG'))['token'])")

mcp() { # mcp '<json body>'
  curl -s -X POST "http://127.0.0.1:$PORT/mcp" \
    -H "Authorization: Bearer $TOK" \
    -H 'Content-Type: application/json' \
    ${SID:+-H "Mcp-Session-Id: $SID"} \
    -d "$1"
}
```

HANDSHAKE AND CAPTURE THE SESSION ID

```
SID=$(curl -si -X POST "http://127.0.0.1:$PORT/mcp" \
  -H "Authorization: Bearer $TOK" -H 'Content-Type: application/json' \
  -d '{"jsonrpc":"2.0","id":1,"method":"initialize","params":{"
    "protocolVersion":"2025-06-18","capabilities":{},
    "clientInfo":{"name":"curl","version":"1"}}}' \
  | awk -F: ' 'tolower($1)=="mcp-session-id"{print $2}' | tr -d '\r')
echo "session: $SID"
```

EVERYDAY CALLS

```
# what tools does this build expose right now?
mcp '{"jsonrpc":"2.0","id":2,"method":"tools/list"}'

# the pack library
mcp '{"jsonrpc":"2.0","id":3,"method":"tools/call",
  "params":{"name":"list_modpacks","arguments":{}}}'

# why did it crash?
mcp '{"jsonrpc":"2.0","id":4,"method":"tools/call",
  "params":{"name":"diagnose_launch_failure",
    "arguments":{"friendlyName":"All the Mods 9"}}}'

# what can be read repeatedly?
mcp '{"jsonrpc":"2.0","id":5,"method":"resources/list"}'
mcp '{"jsonrpc":"2.0","id":6,"method":"resources/templates/list"}'

# liveness, valid even before the handshake
mcp '{"jsonrpc":"2.0","id":7,"method":"ping"}'
```

CONFIRMING THE ADMISSION CHECKS STILL BITE

```
H="Content-Type: application/json"
B='{"jsonrpc":"2.0","id":9,"method":"ping"}'
U="http://127.0.0.1:$PORT/mcp"

curl -so /dev/null -w '%{http_code} no token\n' -X POST "$U" -H "$H" -d "$B"
curl -so /dev/null -w '%{http_code} bad token\n' -X POST "$U" -H "$H" -H 'Authorization: Bearer
nope' -d "$B"
curl -so /dev/null -w '%{http_code} evil origin\n' -X POST "$U" -H "$H" -H "Authorization: Bearer
$TOK" -H 'Origin: http://evil.example' -d "$B"
curl -so /dev/null -w '%{http_code} null origin\n' -X POST "$U" -H "$H" -H "Authorization: Bearer
$TOK" -H 'Origin: null' -d "$B"
curl -so /dev/null -w '%{http_code} GET\n' -X GET "$U" -H "$H" -H "Authorization: Bearer
$TOK"
curl -so /dev/null -w '%{http_code} text/plain\n' -X POST "$U" -H 'Content-Type: text/plain' -H
"Authorization: Bearer $TOK" -d "$B"

# expected: 401 401 403 403 405 415
```

CONFIRMING THE RESOURCE GATE

```
# traversal attempt -- expect -32002 No such resource
mcp '{"jsonrpc":"2.0","id":10,"method":"resources/read",
    "params":{"uri":"mica://modpack/%2e%2e/manifest"}}'
```

C Appendix — source map

Where to look when something in this document needs checking against the code. Paths are relative to `src/main/java/com/micatechnologies/minecraft/launcher/`.

FILE	LINES	READ IT FOR
<code>mcp/McpBootstrap.java</code>	366	Whether the server starts, what gets registered, token and policy plumbing.
<code>mcp/McpServer.java</code>	352	Bind, endpoint file, session lookup, dispatch serialization.
<code>mcp/McpRequestHandler.java</code>	374	Every method, every error code, the order of the gates in a tool call.
<code>mcp/McpClientSetup.java</code>	369	The exact snippets Settings generates, and their quoting rules.
<code>mcp/McpEndpointFile.java</code>	228	File format, permissions, stale-pid handling.
<code>mcp/FxConsentPrompt.java</code>	233	Dialog wording, argument summarisation, the 120 s deadline.
<code>mcp/McpAccessToken.java</code>	157	Token generation, validation, masking.
<code>mcp/McpActivityLog.java</code>	162	What is recorded and how it is redacted.
<code>mcp/transport/LoopbackHttpTransport.java</code>	344	The listener, status codes, body limits.
<code>mcp/transport/McpHttpGuard.java</code>	306	The admission table — the clearest single file in the package.
<code>mcp/transport/StdioProxy.java</code>	218	The relay, and what it does when the launcher is gone.
<code>mcp/approval/McpApprovalEngine.java</code>	158	The decision order, as a pure function.
<code>mcp/approval/LauncherMcpAuthorizer.java</code>	226	How that decision is wired to settings, grants and the prompt.
<code>mcp/approval/McpToolPolicyStore.java</code>	262	Policy persistence and its fail-safe parsing.
<code>mcp/approval/McpGrantStore.java</code>	223	Session grants and their TTL.
<code>mcp/tools/ReadOnlyTools.java</code>	444	The six read-only tools, schemas and output shapes.
<code>mcp/tools/MutatingTools.java</code>	733	The eight state-changing tools, their pre-approval gates and consent framing.
<code>mcp/tools/McpToolResult.java</code>	188	The redaction boundary.
<code>mcp/tools/LiveMcpLauncherView.java</code>	395	What the tools can actually see.
<code>mcp/tools/LiveMcpLauncherActions.java</code>	469	What they can actually do, and the authored-manifest rule.
<code>mcp/resources/McpResourceUri.java</code>	203	The <code>mica://</code> containment gate.

FILE	LINES	READ IT FOR
mcp/resources/LauncherResources.java	219	The three resources and their enumeration.
utilities/SensitiveDataRedactor.java	—	redact vs redactStrict, and which patterns each applies.
LauncherCore.java :224	—	The --mcp dispatch, before the single-instance lock.
LauncherSession.java :318	—	Where the server is started, and why only in the GUI path.
gui/MCLauncherSettingsGui.java	—	The Settings pane's behaviour; resources/gui/settingsGUI.fxml for its layout.

Tests live under `src/test/java/.../mcp/`, one class per production class, and are the fastest way to confirm a behaviour described here without starting the launcher.

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